W3C TPAC 2026 · BREAKOUT SESSIONS

Agentic Web: Standards Landscape

15 breakout proposals, organized into 5 work themes — from definition and UX to permissions, WebMCP, data formats and platform APIs.

Source: github.com/w3c/tpac2026-breakouts (issues #5, #19, #22–#25, #31, #44, #49, #59, #69, #71, #72, #84, #86)

OVERVIEW

15 breakouts cluster into five work themes

01	**Agentic Web Foundations & UX** Definition, future UX, declarative vs imperative, AI-friendly pages	4 sessions
02	**Agent Actions, Permissions & Safety** Permissions/alignment, agent security, consent & high-impact actions, ARIA & risk markup	4 sessions
03	**WebMCP & Tool Discovery** WebMCP + WebExtensions integration for standardized tool calling	1 session
04	**Data Formats, Autofill & Generative UI** Data for humans and AI (JSON-LD/YAML-LD/CBOR-LD), Autofill CG year one, generative UI	3 sessions
05	**Platform APIs, Privacy & On-device Compute** Local capabilities security, Web Speech API, on-device AI model location control	3 sessions

THEME 01 · FOUNDATIONS & UX

Defining the Agentic Web: scope, UX, and interaction patterns

Issue	Session	Host	Focus
#22	**Agentic Web: Definition https://github.com/w3c/tpac2026-breakouts/issues/22**	Apple	What the Agentic Web is; how the Web platform should support AI agents; align on future W3C standardization directions.
#31	**Agentic Web: Future UX https://github.com/w3c/tpac2026-breakouts/issues/31**	Microsoft	Real user scenarios and browser prototypes; prioritize platform tech; seed collaboration projects to incubate key proposals.
#24	**Declarative vs Imperative https://github.com/w3c/tpac2026-breakouts/issues/24**	Apple	How agents interact with existing pages; trade-offs of declarative vs imperative paths; preserve HTML compatibility, a11y, privacy and non-agent users.
#23	**AI-friendly Web Interactions https://github.com/w3c/tpac2026-breakouts/issues/23**	Meta	What makes a page AI-friendly; WebMCP, JSON-LD, existing HTML attributes and the accessibility tree as low-cost augmentation.

Read-out: Apple drives definition + interaction model; Microsoft pushes UX prototypes; Meta focuses on low-cost page-side enablement.

THEME 02 · ACTIONS, PERMISSIONS & SAFETY

Consent, guardrails and risk markup for agent actions

Issue	Session	Host	Focus
#25	**Permissions, Actions & Alignment** https://github.com/w3c/tpac2026-breakouts/issues/25	Apple	Agents acting for users may diverge from user intent/values; design browser-side guardrails, permission management and safeguards against accidental or harmful actions.
#5	**Agent Security Considerations** https://github.com/w3c/tpac2026-breakouts/issues/5	Google	Security risks introduced by AI agents; gather community feedback on Web platform security proposals; extend existing Web security mechanisms to new agent-driven risks.
#71	**Consent, High-Impact Ops, Neutrality** https://github.com/w3c/tpac2026-breakouts/issues/71	Google	Browser mediates permissions/liability among user, agent and logged-in site; consent UX, high-consequence operations, agent neutrality, security, privacy, identity and competition fairness.
#69	**ARIA Protection & Risk Markup** https://github.com/w3c/tpac2026-breakouts/issues/69	Google	Need for explicit action semantics for agents; prevent abuse of ARIA for machine consumption; mark risk level in HTML so agents grasp buttons, links and inputs and their impact.

Read-out: Google owns 3 of 4 sessions here — consent model, security architecture, and HTML-level risk semantics are the battleground.

THEME 03 · WEBMCP & TOOL DISCOVERY

WebMCP meets WebExtensions: declared tools over scraping

Issue #84 · Hosts: Intel, Apple, and former Google engineers

THE PROBLEM

Today agents drive pages by parsing the DOM, reading screenshots, or invoking automation scripts — fragile, privacy-invasive, and controlled by the agent rather than the site.

THE PROPOSAL

Integrate WebMCP with browser extensions so sites explicitly declare structured tools; agents discover and call them through a standard channel, with the site retaining control and consent.

https://github.com/w3c/tpac2026-breakouts/issues/84

WHY IT MATTERS

Declared, not scraped
Site publishes its own tool schema

Structured & auditable
Tool calls are machine-readable and reviewable

Controllable by the site
Fits permissions model from Theme 02

THEME 04 · DATA FORMATS, AUTOFILL & GENERATIVE UI

Data for humans and AI, and UI agents can render

Issue	Session	Host	Focus
#86	**Data Formats for Humans and AI https://github.com/w3c/tpac2026-breakouts/issues/86**	W3C	Interoperable structured data for humans and AI; JSON-LD, YAML-LD (readable) and CBOR-LD (compact, constrained environments).
#19	**Autofill CG: Year One & Roadmap https://github.com/w3c/tpac2026-breakouts/issues/19**	Shopify	Review first year of Autofill Community Group; experimental results, practitioner feedback, next steps and standardization plan.
#44	**Generative UI: Optimization & Monitoring https://github.com/w3c/tpac2026-breakouts/issues/44**	Huawei	Web-engine rendering is flexible but heavy; JSON-native rendering is light but layout-limited; seek a standard balancing flexibility, performance and interop.

Read-out: W3C drives the data layer (JSON-LD family); Shopify owns the practical Autofill track; Huawei raises the generative-UI performance question — directly relevant to our immersive-media and on-device rendering work.

THEME 05 · PLATFORM APIs, PRIVACY & ON-DEVICE COMPUTE

Exposing more device capability while keeping users in control

Issue	Session	Host	Focus
#59	**Local Capabilities APIs: Security & Privacy https://github.com/w3c/tpac2026-breakouts/issues/59**	Google	WebBluetooth / WebUSB / WebNFC / WebSerial / WebHID: balance opening device access with user control and platform security.
#49	**Speech Recognition & Interaction API https://github.com/w3c/tpac2026-breakouts/issues/49**	Google, Firefox	Web Speech API update; on-device recognition for privacy, lower latency and cross-browser interop; next-gen voice interaction.
#72	**On-device AI: User Compute-Location Control https://github.com/w3c/tpac2026-breakouts/issues/72**	Samsung	Let users control where in-browser AI models run; resolve conflicts with existing privacy settings; unified compute-location identifier.

Read-out: Google owns device APIs (#59) and speech (#49); Samsung raises the compute-location question (#72) — a direct hook for on-device / edge-agent architecture and our 6G-edge agenda.

TAKEAWAYS & RECOMMENDATIONS

Where to focus: scope, safety, WebMCP, edge compute

1 **Apple drives the definition and interaction model** (#22, #24, #25); Google drives safety and action semantics (#5, #69, #71). Meta and Microsoft cover developer-side enablement and UX.

2 **WebMCP + WebExtensions (#84)** is the most concrete interop bet; expect it to anchor future W3C work on agent tool calling. Attend and track the spec direction.

3 **Consent + risk-markup (#71, #69) is the next two years** of W3C agent work. Protecting ARIA from misuse is a direct accessibility concern — prepare a position.

4 **Aligned with our agenda:** on-device compute location (#72), CBOR-LD compact data (#86), and generative-UI rendering (#44) map directly to 6G edge agents and immersive media.

Recommended actions: observe #22 / #24 / #84 / #69 / #72 as guests; co-author a note linking Agentic Web to AIDN and trusted identity; file a position on ARIA-vs-agent semantics.